

Middleware: KAA (<http://www.kaaproject.org/#2>) and OpenRemote (<http://www.openremote.com/>) are open source projects, which provide an end point SDK that can be embedded into connected devices. They support both full OSs and OS-less microchips alike, with virtually any hardware from low-powered MCUPs.

Operating system: Snappy Ubuntu Core (<http://developer.ubuntu.com/en/snappy/>), Contiki (<http://www.contiki-os.org/>) and RIOT (<http://www.riot-os.org/>) are the most preferred OSs. Snappy Ubuntu Core offers a lightweight and more reliable OS for IoT based applications. Contiki provides a full-fledged OS dedicated for WSN with underlying support for various hardware devices. RIOT is another similar IoT-friendly OS designed to minimise the memory footprint and enhance energy efficiency. Extensive community support leads to continuous evolution of these tools.

Open source in Big Data

Big Data has become a reality now, and it is important for the business continuity. The volume, variety, velocity and veracity in which data is accumulated by organisations has led to phenomenal growth not just in understanding Big Data but also better and faster analytics. Due to its popularity, a lot of open source projects have evolved around Big Data and some of the most popular ones listed below serve different purposes.

- **Druidd** is an open source distributed data store originally developed to analyse online events for ad markets and to work on data streams.
- **Apache Spark**: This is a newer data processing engine used to run analysis faster on large datasets. It has support for applications written in Java, Scala, Python and R.
- **Apache Flume**: This is a service that gathers information from distributed sources that is later stored in HDFS.
- **Apache Hive**: This tool allows people to use an SQL-like language to analyse petabytes of data.
- **Taiga**: Originally developed by LinkedIn, this tool is

responsible for distributed stream processing.

- **HBase**: This open source NoSQL database has been designed for backend search engines and has the ability of returning search results faster.

Open source in containers

Container technology has changed the way in which applications are deployed and managed. Containers provide the capability to host different isolated applications using resource isolation features of the OS.

Docker is the most popular open source container technology due to its light weight and the ability to package applications with all dependencies into standardised units of software development. With Docker, the developer doesn't need to waste hours in building a dev environment, nor make copies of production code and spin up new instances. Neither does the developer need to eliminate the environment inconsistency by packaging an application in such a way that it can run on any machine, either for test or production. This avoids the overhead of installing configs on different systems. Docker creates a common framework for developers and sysadmins to work together on distributed applications.

The evolution of open source is a continuous process as many developers are continually working on enhancing the features. Sooner or later, open source solutions might become the de-facto standard for applications in many fields as they are being adopted by academia and enterprise alike. 

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Torch

Torch is a framework for scientific computing. It has great support for many machine learning algorithms. It is based on a faster scripting language called LuaJIT, C and CUDA implementations. The primary features of Torch are:

- Neural network and energy based models
- Quick and efficient GPU support
- It has embeddable ports for iOS and Android
- Robust and powerful N-dimensional array handling
- Many functions for indexing, slicing and transposing
- Comes with a large ecosystem of community-driven packages for computer vision, machine learning, etc

Apart from the software libraries highlighted in this article, there are various other options for deep learning development like Deepnet, Hebel, Deeppy, Neon, Intel Deep learning framework and Mocha. As the field of deep learning is in its

early stages and the best is yet to come, it can be confidently said that in the years ahead, deep learning will emerge as a game changer with applications in various domains. 

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